Coefficient of contraction, C_C =

$$\frac{area\ of\ jet\ at\ venacontracta\ (a_c)}{area\ of\ Orifice\ (a)}$$

i.e., $C_C = \frac{a_c}{a}$

2. Coefficient of velocity

The ratio of velocity of jet at vena contracta to the theoretical velocity is termed as coefficient of velocity. It is represented by C_V.

Coefficient of velocity, C_V =

$$\frac{actual\ velocity\ of\ jet\ at\ venacontracta\ (V_a)}{theoretical\ velocity\ (V_t)}$$

i.e., $C_V = \frac{V_a}{V_t}$

3. Coefficient of discharge

The ratio of actual discharge through the Orifice to the theoretical discharge is termed as coefficient of discharge. It is denoted by C_d.

Coefficient of discharge, C_d =

$$\frac{actual\ discharge\ (Q_a)}{theoritical\ discharge\ (Q_t)}$$

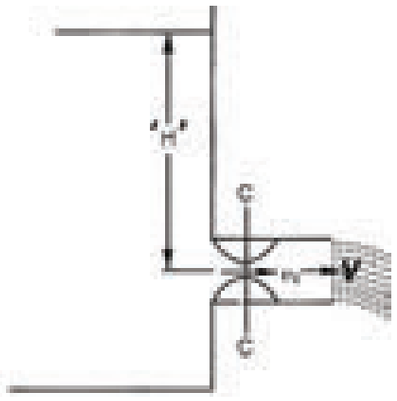
i.e., $Cd = \frac{Q_a}{Q_t}$

7.4.6 Practical applications of Orifice

1. Circular, square and rectangular sluices provided in irrigation tanks.
2. Vent ways of culverts.
3. The openings provided in swimming pools.
4. The openings of canal lock gates to fill up and empty.
5. The openings of balancing reservoirs and surge tanks.

7.4.7 Mouthpiece – Definition

A mouthpiece is a short length of pipe fitted to an Orifice either inwardly or outwardly. Generally, the length of the pipe varies from twice to thrice the diameter of Orifice. It is fitted to increase the discharge of the Orifice.



V_C = Velocity at Venacontracta

V = Velocity at Mouth piece



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